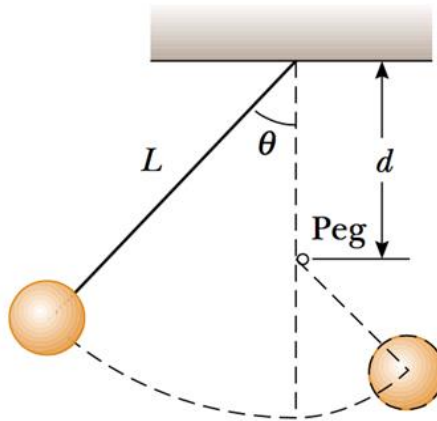


## Tutorial 3

### Equilibrium, Work Done, Energy and Power

#### 1 Peg and Pendulum

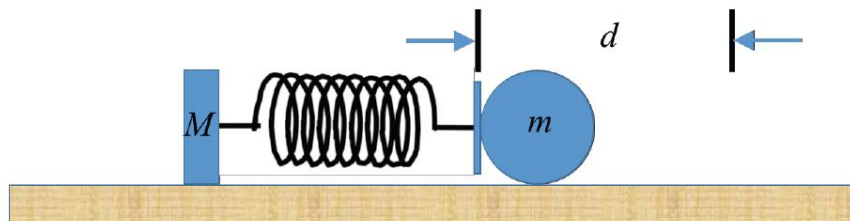
A pendulum, comprising a string of length  $L$  and a small sphere swing in the vertical plane. The string hits a peg located a distance  $d$  below the point of suspension. Show that if the pendulum is released from an angle  $\theta$  from the vertical as shown in the diagram below, and is to swing in a complete circle centred on the peg, then the minimum value of  $d$  is  $(3 + 2\cos\theta)\frac{L}{5}$



Show also that for all values of  $\theta$ , the sphere will not hit the ceiling should it complete the circular motion.

#### 2 Sphere losing contact with the spring.

As shown in Figure 3, a spring of spring constant  $k$  is placed horizontally on the floor. One end of the spring is attached to a solid block of mass  $M$  and the other end of the spring is compressed by a sphere of mass  $m$  by a distance  $d$ . Neglect all the friction and neglect the mass of the spring as well as the mass of the plate between the sphere and the spring.



**Figure 3**

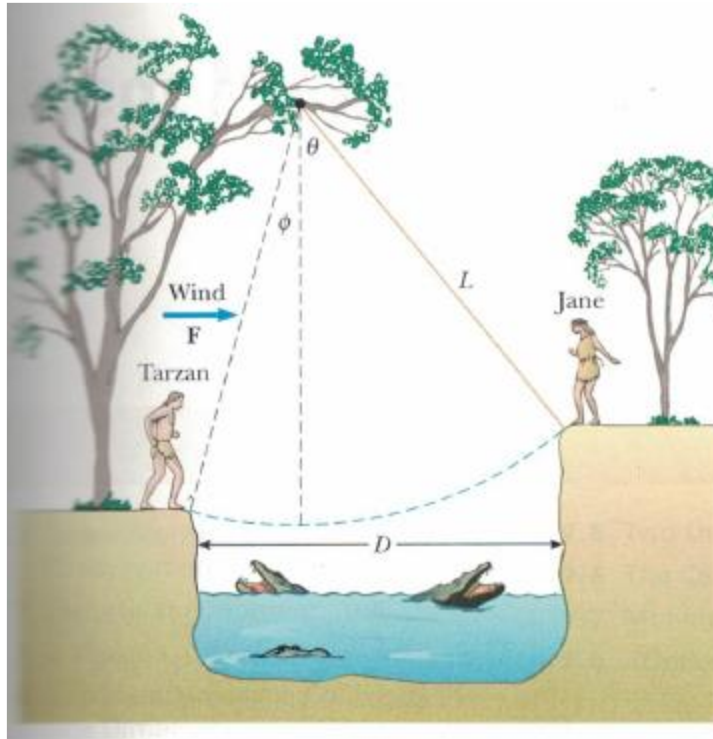
- The block is fixed on the floor. Derive an expression for the speed of the sphere when it is at a distance  $d$  from the initial position.
- The block is now free to move, derive an expression for the speed of the sphere relative to the floor after it loses the contact with the spring.
- The block is still free to move. Derive an expression for the distance the block travels when the sphere just loses the contact with the spring.

**Answers:**

$$(a) \sqrt{\frac{k}{m}} d \quad (b) \sqrt{\frac{M}{m(m+M)}} kd \quad (c) \left(\frac{m}{m+M}\right) d$$

### 3 Tarzan and Jane.

Jane whose mass is 50.0 kg, needs to swing across a river having width  $D$  filled with man-eating crocodiles to save Tarzan from danger. However, she must swing into a wind exerting constant horizontal  $\mathbf{F}$  on a vine having length  $L$  and initially making an angle  $\theta$  with the vertical as shown in the figure below. Taking  $D = 50.0$  m,  $F = 110$  N,  $L = 40.0$  m and  $\theta = 50.0^\circ$ ,

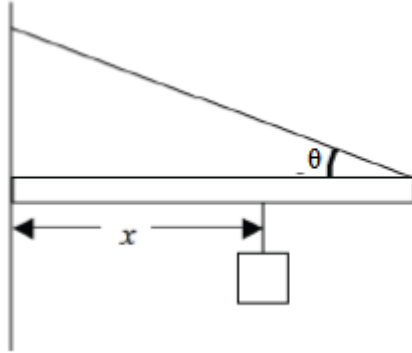


- (a) With what minimum speed must Jane begin her swing to just make it to the other side?
- (b) Once the rescue is complete, Tarzan and Jane must swing back across the river. With what minimum speed must they begin their swing? Assume that Tarzan has a mass of 80.0 kg.

**Answer**      (a)    6.15 m/s      (b)    9.87 m/s

#### 4 Block hanging on a stick in equilibrium.

One end of an uniform stick of mass  $M$  and length  $l$  is placed against a vertical wall. The other end is held by a lightweight cord that makes an angle  $\theta$  with the stick as shown in the figure below. The coefficient of static friction between the end of the stick and the wall is  $\mu_s$ . A block of mass  $m$  is suspended from the stick at a distance  $x$  from the wall.



Show that the minimum value of  $x$  for which the stick can remain in equilibrium is given by:

$$x_{\min} = \left[ \frac{1 + \frac{M}{2m}(1 - \mu_s \cot \theta)}{1 + \mu_s \cot \theta} \right] l$$

We need  $0 \leq x_{\min} \leq l$ . Explain what it means when the  $x_{\min} < 0$  or when the  $x_{\min} > l$ . Show also that the system cannot be equilibrium if there is no friction between the wall and the stick.

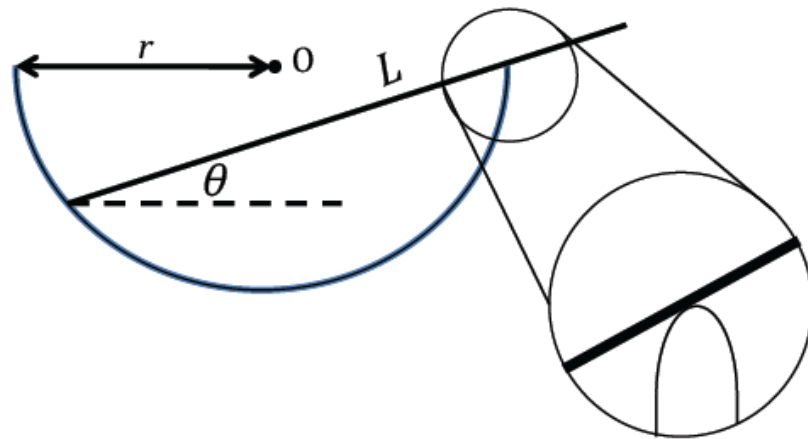
#### Answer

$x_{\min} < 0$  means the stick will always be balanced.

$x_{\min} > l$  means the stick can never be balanced.

## 5. Equilibrium – Contact Forces

A friend gives you a uniform, smooth and thin rod of length  $L$  and weight  $W$ , and challenges you to find an angle  $\theta$  (the angle the rod makes with the horizontal) such that the rod will rest in equilibrium on the rounded rim of the hemispherical bowl with centre  $O$  and radius  $r$ .

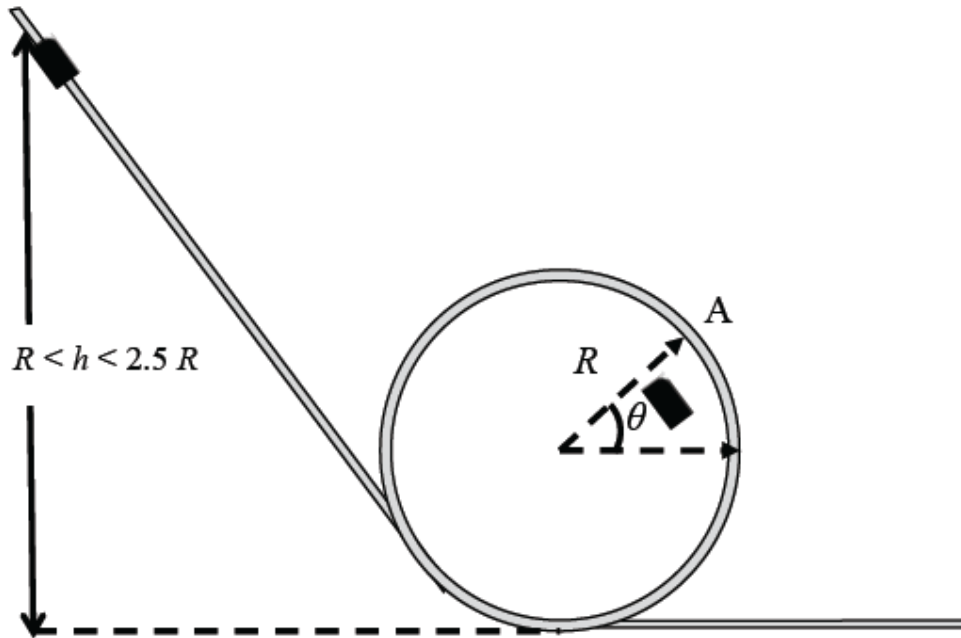


Show that  $4r\cos 2\theta = L\cos\theta$ .

## 6. Conservation of Energy

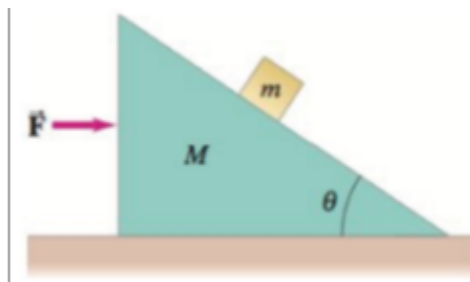
As shown in Figure 1(a), a block is released at height  $h$  where  $h$  is  $R < h < 2.5 R$ . The block slides down the frictionless track and up the vertical circular loop. The block loses contact with the track at position A as shown in Figure 1. Show that

$$\sin \theta = \frac{2(h-R)}{3R}.$$



## 7. Block on a moving slope

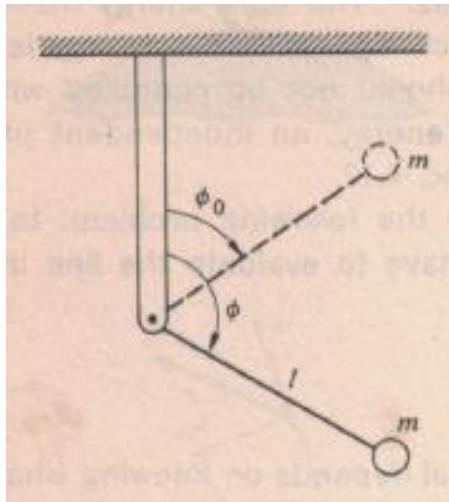
A small block of mass  $m$  rests on the sloping side of a triangular block of mass  $M$  which itself rests on a horizontal table as shown in the figure. Assuming the surface between the triangular block and the table is smooth while the surface between the two blocks are rough with coefficient of static friction  $\mu_s$ . Determine the range of  $F$  that can be applied to mass  $m$  so that it will not slide down the slope.



**Answer:** 
$$\frac{\tan \theta - \mu_s}{1 + \mu_s \tan \theta} (m + M)g \leq F \leq \frac{\tan \theta + \mu_s}{1 - \mu_s \tan \theta} (m + M)g$$

## 8. The Inverted Pendulum

A pendulum consists of a light rigid rod of length  $l$ , pivoted at one end and with mass  $m$  attached at the other end. The pendulum is released from rest at angle  $\phi_0$  as shown. What is the velocity of  $m$  when the rod is at angle  $\phi$ ?



What is the maximum velocity and how do we achieve it?

**Answer:**  $v(\phi) = \sqrt{2gl(\cos\phi_0 - \cos\phi)}$

## 9. Equivalent system of forces

A force system acting on a rigid body consists of three forces

$$\begin{aligned}\mathbf{F}_1 &= \mathbf{i} - \mathbf{j} + 2\mathbf{k} \\ \mathbf{F}_2 &= \mathbf{i} + 3\mathbf{j} - \mathbf{k} \\ \mathbf{F}_3 &= s\mathbf{i} + t\mathbf{j} + 2\mathbf{k}\end{aligned}$$

Where  $s$  and  $t$  are constants. The forces act respectively through the points whose position vectors relative to origin  $O$  are

$$3\mathbf{i} - \mathbf{j} + \mathbf{k}, \quad \mathbf{j} + 2\mathbf{k}, \quad \mathbf{k}.$$

Obtain in terms of  $s$  and  $t$  the equivalent force system consisting of a single force  $\mathbf{F}$  acting through  $O$  and a couple of moment  $\mathbf{G}$ . Determine  $s$  and  $t$  so that  $\mathbf{G}$  is parallel to  $\mathbf{F}$ .

**Answer:**

$$\mathbf{F} = (2 + s)\mathbf{i} + (2 + t)\mathbf{j} + 3\mathbf{k}, \quad \mathbf{G} = (-8 - t)\mathbf{i} + (s - 3)\mathbf{j} - 3\mathbf{k}; \quad s = \frac{7}{2}, \quad t = -\frac{5}{2}$$

### 10. Book sliding on a rough surface

A 2.50 kg textbook is forced against a horizontal spring of negligible mass and force constant 250 N/m, compressing the spring a distance of 0.250 m. When released, the textbook slides on a horizontal tabletop with coefficient of kinetic friction  $\mu_k = 0.30$ . Use work-energy theorem to find how far the textbook moves from its initial position before coming to rest.

**Answer :** 1.1 m

### 11. Cycling up and down hill with air resistance

A bicyclist of mass 75 kg (including the bicycle) can coast down a  $4.0^\circ$  hill at a steady speed of 12 km/h. Pumping hard, the cyclist can descend the hill at a speed of 32 km/h. Using the same power, at what speed can the cyclist climb the same hill? Assume the force of friction is proportional to the square of the speed.

**Answer :** 29 km/h

### 12. Pushing a crate up a rough ramp

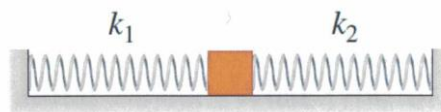
A 20.0 kg crate sits at rest at the bottom of a 15.0 m long ramp that is inclined at  $34.0^\circ$  above the horizontal. A constant horizontal force of 290 N is applied to the crate to push it up the ramp. While the crate is moving, the ramp exerts a constant frictional force on it that has magnitude 65.0 N.

- (a) What is the total work done on the crate during its motion from the bottom to the top of the ramp?
- (b) How much time does it take the crate to travel to the top of the ramp?

**Answer :** (a) 987 J, (b) 3.02 s

### 13. Coupled Springs

A 3 kg block is connected to two ideal horizontal springs having force constants  $k_1 = 25$  N/cm and  $k_2 = 20$  N/cm. The system is initially in equilibrium on a horizontal, frictionless surface. The block is now pushed 15 cm to the right and released from rest.



- (a) What is the maximum speed of the block? Where in the motion does the maximum speed occur?
- (b) What is the maximum compression of spring 1?

**Answer :** (a) 5.81 m/s (b) 15 cm

#### 14. Uniform rod tied to an elastic string

A uniform rod AB of mass  $m$  and length  $2l$  is smoothly jointed at A to a fixed point. A light elastic string of length  $l$  and modulus of elasticity  $mg$  connects the end B to a point C distant  $2l$  vertically above A. Show that there are two positions in which the rod can rest in equilibrium with the string taut and determine whether the equilibrium at these positions is stable or unstable.

**Note:** An elastic string of length  $l$  and modulus of elasticity  $\lambda$  when stretch with an extension  $x$  has tension

$$T = \lambda \frac{x}{l}$$

